



Multi-scale characterisation and damage analysis of 3D braided composites under off-axis tensile loading

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ARTICLE INFO

Keywords:

- A. Textile composites
- B. Mechanical properties
- B. Fracture
- C. Representative volume element
- C. Damage mechanics

ABSTRACT

This study investigates the influence of off-axis tensile loading (0°, 45°, 60° and 90°) on the mechanical behavior of 3D braided composites. Here, digital image correlation is utilized to characterize the full-field deformation and strain distribution in the composites. Scanning electron microscope and micro-computed tomography techniques are also employed to study the deformation and damage mechanisms in the fractured specimens. An off-axis stiffness prediction model for 3D braided composites is proposed, and a finite element model of braided composites containing porosity is established. A user-defined material subroutine is also developed to implement the damage model as a function of off-axis loading angle. The experimental and simulation results demonstrate that the application of off-axis loading causes significant variations in the spatial orientation angle (γ , φ) of the yarn. This phenomenon gives rise to two distinct stages in which the modulus of the composite initially increases, followed by a subsequent decrease. Furthermore, the dominant failure mechanisms in these on-axis samples are found to be yarn breakage and matrix cracking. Also, matrix cracks and filaments pull-out are observed in the off-axis samples.

Nomenclature

a	the semi-major axis length of the yarn cross-sectional parameters.
L	the pitch length
H	the pitch height
γ	the interior braiding angle (°)
σ	stress
E	tension moduli
ν	Poisson's ratio
T	tensile strength
S	shear strength
φ	the angle between the yarn's projection on the yoz plane and the y-axis
$[\bar{S}]$	flexibility matrix
ρ	the mixing coefficient
b	the semi-minor axis length of the yarn cross-sectional parameters.
W	the pitch width
V_y	the yarn volume fraction
ϵ	strain

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l_i, m_i, n_i ($i = 1, 2, 3$)	the direction cosines of the angles between the off-axis and the on-axis coordinate system.
G	shear moduli
V	volume fractions
C	compressive strength
d_i	damage variable
γ'	the off-axis braided angle
$[\bar{C}]$	stiffness matrix
θ	the off-axis angle

1. Introduction

Due to the excellent interlaminar fracture toughness, impact resistance and damage tolerance of 3D braided composites over their 2D counterparts [1–3], 3D braided composites offer great potential for structural applications in the aerospace industry [4]. However, such materials are inevitably subjected to off-axis loads [5,6], which may cause damage and ultimately the failure of the structure. Given that their

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<https://doi.org/10.1016/j.compscitech.2024.111017>

Received 1 August 2024; Received in revised form 5 December 2024; Accepted 14 December 2024

Available online 16 December 2024

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complex nature includes a highly inhomogeneous microstructure which involves yarns, pores and other inclusions [7,8], the testing and analysis of 3D braided composites poses a major challenge. Therefore, it is necessary to study the mechanical properties and failure mechanisms in such materials under off-axis loading in order to optimize their design of and improve the safety of structures based on such materials.

Off-axis loading alters the damage process in a material by subjecting it to a combination of normal and shear stresses, thereby affecting its structural integrity and reliability [9,10]. Previous studies have shown that the mechanical response of a composite is highly dependent on the loading angle, since the local stress state varies with off-axis angle [11–15]. Teng et al. [16] studied the deformation response and failure behavior of 2D woven C/SiC composites subjected to on-axis and off-axis tensile loading. Their test results indicate that the materials exhibit failure modes dependent on the off-axis angle. Guan et al. [17] investigated the in-plane compressive behavior of a 2D plain-woven composite using on-axis and off-axis compression experiments under both quasi-static and dynamic loading conditions. Altaf et al. [18] concluded that delamination and yarn rotation were the dominant failure mechanisms during off-axis loading of 2.5D C/SiC composites. The above research focused on the mechanical properties of 2D composite materials under off-axis loading without considering the effects of internal defects on mechanical behavior. Due to the complex spatial structure and unknown damage mechanisms in 3D braided C/SiC composites when subjected to off-axis loading, it is necessary to undertake relevant experimental studies.

Clearly, 3D braided C/SiC composites offer intrinsic multiscale properties. Numerical modelling has been used for predicting their mechanical properties and damage evolution [19]. Zhu et al. [20] investigated the effect of braiding parameters and loading direction on the compressive behavior and failure mechanisms in 3D braided composites at the mesoscale. Du et al. [21] studied tensile damage development and the strength of 3D five-directional(5d) braided composites using a parametric finite element model. They showed that the FE model could effectively predict both the strength and damage in 3D5d braided composites. Hu et al. [22] proposed a microstructure model for characterizing transverse impact damage. Xu et al. [23] developed a bridging model to predict the impact response of 3D braided composites. The research showed that the damage process in a composite material can be accurately simulated using finite element simulation techniques using appropriate damage criteria. However, the influence of off-axis loading on the mechanical properties of these materials has not been fully studied, and a suitable model to predict the mechanical properties remains to be developed.

Due to the limited number of investigations on the prediction of their mechanical response under complex loading conditions, 3D braided composite structures are generally designed with a conservative safety margin. As a result, the full potential of these structures has not been achieved. The aim of this study is to investigate the mechanical properties and failure behavior of 3D braided composites under various off-axis load conditions using the DIC technique. Following a theoretical prediction model combined with a Micro-CT based, a multiscale finite element analysis is developed to investigate progressive failure in 3D braided composites under off-axis loading. Micro-CT and 3D reconstruction techniques are applied to detect the distribution of internal porosity. The FE model is developed to explore the damage and failure mechanisms in these 3D braided composites. Provided the theoretical contribution for the composite structure and their application in aerospace construction.

2. Off-axis tensile testing

2.1. Specimen design

The 3D braided C/SiC composite specimens were fabricated by the China Textile Academy. T700-12K (Toray®, Tokyo, Japan) carbon fibers

were used to fabricate 3D braided preforms via a four-step process [24]. The braided preforms were impregnated with a polycarbosilane (PCS)-xylene solution over 9–12 cycles, and then pyrolyzed in an inert atmosphere at 1473 K [25–27]. The specimen design, shown in Fig. 1, was used for conducting tensile tests on braided composites. The sample has a width of 12 mm, with a length of 130 mm. Specimens were cut at angles of 0°, 45°, 60°, and 90°. Additionally, the rectangular specimens with a constant width of 10 mm and a gauge length of 25 mm were designed following the general guidelines of JC/T 2404-2017 [28], which specifies the standard test methods for tensile properties of ceramic matrix composite materials.

2.2. Tensile testing using the Micro-CT and DIC method

Micro-CT imaging was performed using the Xradia 610 Versa system with a $20 \times$ lens detector. This non-destructive method allows for the visualization and evaluation of the internal structural characteristics of a material using an X-ray source [29–31]. The sample underwent a 360° rotation, while capturing images. The resolution for microscopic scanning is 2 μ m. Fig. 2a shows the scanned zone. Image segmentation was conducted to differentiate the yarn, matrix, and pores. For segmentation, Dragonfly software (Version 2022) was employed, utilizing its supervised machine learning tools. This feature allowed for the classification of different material phases by training models based on manually labeled regions. To mitigate the presence of image noise, a 3D median filter with a kernel size of $3 \times 3 \times 3$ was applied. Following segmentation, the images were reconstructed in 3D using Dragonfly's Volume Rendering module, enabling precise visualization and quantitative analysis of the internal structure, including pore size, yarn orientation, and fiber distribution in matrix.

Fig. 2b shows the experimental configuration used for conducting the tensile tests. All tests were conducted on an MTS-880/25 electro-mechanical test machine with a maximum load capacity of 250 kN and a stroke of ± 50 mm. The displacement rate of the upper crosshead was set to 2 mm/min, following the JC/T 2404-2017 testing Standard [28]. The experimental setup is illustrated in Fig. 2b. Three samples were tested to obtain an average value. Before applying a speckle pattern, a background color was sprayed onto the surface of the specimen [32–34]. In order to accurately capture the representative behavior of the sample

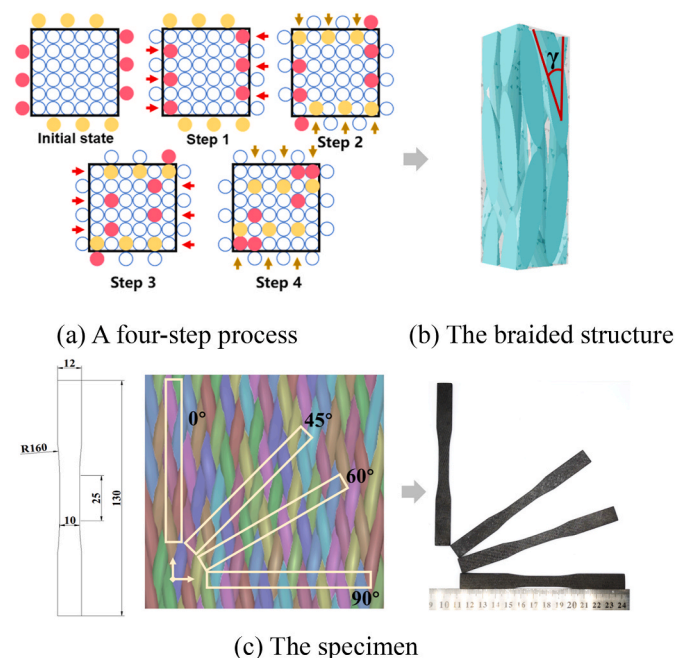


Fig. 1. The specimen design.

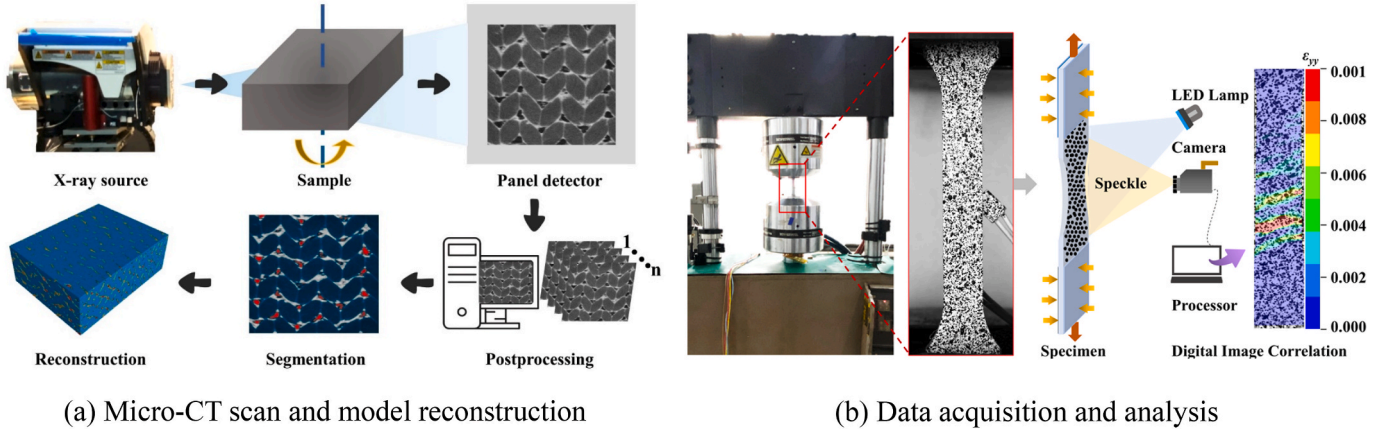


Fig. 2. Experimental setup and procedures.

using this technique, it is imperative that the speckle pattern encompasses a sufficient quantity of black speckles, exhibiting diverse shapes and sizes [35]. The DIC system includes a high-resolution camera and LED lighting to capture strain distribution during the tensile tests.

3. Finite element analyses

3.1. Representative volume element (RVE) reconstruction based on Micro-CT

The mesoscopic parameters obtained in this part of the study were used to establish the finite element models. It is possible to accurately determine the internal geometric parameters of the 3D braided C/SiC composites using a Micro-CT analysis. The yarn is assumed to be elliptical in shape. The yarn cross-sectional parameters (the semi-major axis length a , the semi-minor axis length b) were derived by determining the braid parameters (the pitch length L , the pitch width W , the pitch height H) using individual 2D image slices of the braided sample, as shown in Fig. 3a. These parameters were calculated by averaging at least three RVEs. The yarn volume fraction V_y and the RVE parameters are listed in Table 1. The mesoscopic parameters of the RVE were calculated using the following equations:

$$b = \frac{\sqrt{2}}{8} W \quad (1)$$

$$a = \sqrt{3} b \cos \gamma \quad (2)$$

$$\tan \gamma = \frac{8b}{H} \quad (3)$$

where γ is the internal braiding angle.

According to the 3D reconstruction data, the measured parameters were applied to construct the RVE with a parametric modeling method by using SolidWorks. The finite element analysis was performed using commercial code Abaqus/Standard. Here, the 3D four-node tetrahedral (C3D4) element, with a size of 0.01 mm, was selected to mesh the model. A mesh sensitivity analysis has been reported in a previous study [19]. Fig. 3b presents the meshing results of the FE model. The voids are established by randomly selecting elements using the Monte Carlo method [36], which is a probabilistic technique that employs random sampling to model stochastic processes. In this study, the void elements were assigned with extremely low elastic modulus values ($E = 10^{-10}$ GPa) to represent the mechanical effects of voids within the material.

3.2. Off-axis loading

In the on-axis case, the periodic boundary conditions [37] were established through the application of on-axis displacement loads. The displacements of corresponding nodes on opposite faces of the RVE were coupled to ensure continuity and periodicity. This approach ensures that the deformation field within the RVE remains consistent with the

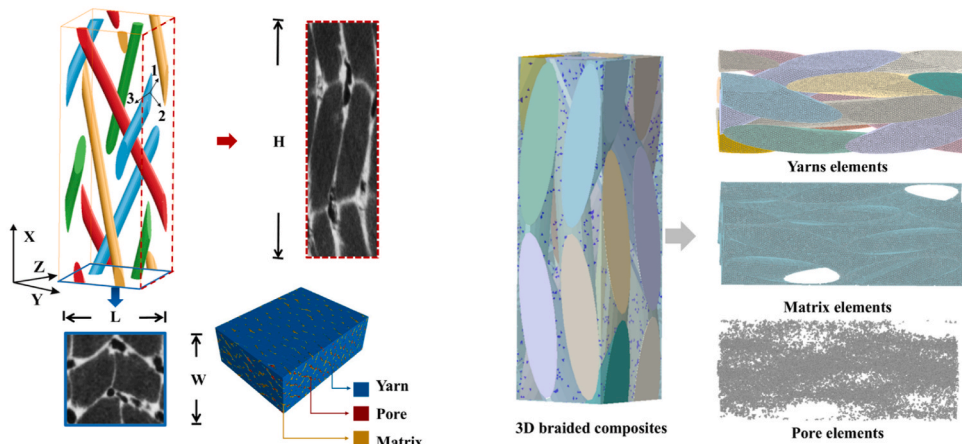


Fig. 3. The acquisition of 3D parameters based on the CT-scan.

Table 1

Modeling parameters for the RVE at the mesoscale.

H (mm)	W (mm)	L (mm)	γ (°)	V_y (%)	Porosity (%)
6.05	1.65	1.63	21.04	66.25	5.78

periodic nature of the composite structure. Similarly, for the case of off-axis loading, 3D braided composites were subjected to loading in both the longitudinal and transverse directions, resulting in the generation of shear stresses within the loading plane. Displacement loading was also used to impose off-axis loads on the RVE model. The strains in the off-axis coordinate system ($\bar{\epsilon}$) can be transformed to the strains in the on-axis coordinate system (ϵ) using Eq. (4):

$$\epsilon = B^{-1}\bar{\epsilon} \quad (4)$$

where for in-plane off-axis loading $\bar{\epsilon}_1 \neq 0, \bar{\epsilon}_2 = \bar{\epsilon}_3 = \bar{\epsilon}_{12} = \bar{\epsilon}_{13} = \bar{\epsilon}_{23} = 0$. The corresponding displacement components in the on-axis coordinate system were obtained and then introduced into on RVE model based on the periodic boundary conditions.

The normal stress and shear stress in each principal direction in the on-axis coordinate system can be obtained from the simulation. Then, the stress in the on-axis coordinate system (σ) can be transformed into the stress in the off-axis coordinate system ($\bar{\sigma}$) using:

$$\bar{\sigma} = A\sigma \quad (5)$$

The spatial stress transformation matrix A and the strain transformation matrix B can be expressed as:

$$[A] = \begin{bmatrix} l_1^2 & m_1^2 & n_1^2 & 2m_1n_1 & 2l_1n_1 & 2l_1m_1 \\ l_2^2 & m_2^2 & n_2^2 & 2m_2n_2 & 2l_2n_2 & 2l_2m_2 \\ l_3^2 & m_3^2 & n_3^2 & 2m_3n_3 & 2l_3n_3 & 2l_3m_3 \\ l_2l_3 & m_2m_3 & n_2n_3 & m_2n_3 + m_3n_2 & l_2n_3 + l_3n_2 & l_2m_3 + l_3m_2 \\ l_3l_1 & m_3m_1 & n_3n_1 & m_3n_1 + m_1n_3 & l_3n_1 + l_1n_3 & l_3m_1 + l_1m_3 \\ l_1l_2 & m_1m_2 & n_1n_2 & m_1n_2 + m_2n_1 & l_1n_2 + l_2n_1 & l_1m_2 + l_2m_1 \end{bmatrix} \quad (6)$$

$$[B] = \begin{bmatrix} l_1^2 & m_1^2 & n_1^2 & m_1n_1 & l_1n_1 & l_1m_1 \\ l_2^2 & m_2^2 & n_2^2 & m_2n_2 & l_2n_2 & l_2m_2 \\ l_3^2 & m_3^2 & n_3^2 & m_3n_3 & l_3n_3 & l_3m_3 \\ 2l_2l_3 & 2m_2m_3 & 2n_2n_3 & m_2n_3 + m_3n_2 & l_2n_3 + l_3n_2 & l_2m_3 + l_3m_2 \\ 2l_3l_1 & 2m_3m_1 & 2n_3n_1 & m_3n_1 + m_1n_3 & l_3n_1 + l_1n_3 & l_3m_1 + l_1m_3 \\ 2l_1l_2 & 2m_1m_2 & 2n_1n_2 & m_1n_2 + m_2n_1 & l_1n_2 + l_2n_1 & l_1m_2 + l_2m_1 \end{bmatrix} \quad (7)$$

where l_i , m_i and n_i ($i = 1, 2, 3$) are the direction cosines of the angles between the off-axis and the on-axis coordinate system.

3.3. Progressive damage model

The moduli and strengths of the yarns can be calculated using the Chamis model [38], as shown in Eqs. (8) and (9).

$$\begin{cases} E_{y11} = V_f E_{f11} + V_m E_m \\ E_{y22} = E_{y33} = \frac{E_m}{1 - \sqrt{V_f}(1 - E_m/E_{f22})} \\ G_{y12} = G_{y13} = \frac{G_m}{1 - \sqrt{V_f}(1 - G_m/G_{f12})} \\ G_{y23} = \frac{G_m}{1 - \sqrt{V_f}(1 - G_m/G_{f23})} \\ \nu_{y12} = \nu_{y13} = V_f \nu_{f12} + V_m \nu_m \\ \nu_{y23} = V_f \nu_{f23} + 2V_m \nu_m - (E_{fb2}/E_{fb1}) V_m \nu_{y12} \end{cases} \quad (8)$$

where E , G , ν and V represent the tension moduli, shear moduli, Poisson's ratio and volume fractions. The subscripts 11, 22, 33, 12, 13, 23 denote the longitudinal, transverse, and normal directions, as well as the respective shear planes in the material. The subscripts y , f and m indicate yarn, carbon fiber, and matrix components, respectively.

$$\begin{cases} T_{y11} = T_f (V_f + V_m E_m/E_{f11}) \\ C_{y11} = C_f (V_f + V_m E_m/E_{f11}) \\ T_{y22} = T_{y33} = (1 - (\sqrt{V_f} - V_f)(1 - E_m/E_{f22})) T_m \\ C_{y22} = C_{y23} = (1 - (\sqrt{V_f} - V_f)(1 - E_m/E_{f22})) T_m \\ S_{y12} = S_{y13} = (1 - (\sqrt{V_f} - V_f)(1 - G_m/G_{f12})) T_m \\ S_{y23} = \sqrt{\frac{1 + T_{y22}/C_{y22}}{T_{y22}C_{y22}3 + 5T_{y22}/C_{y22}}} \end{cases} \quad (9)$$

where T , C , S represent the tensile, compressive, and shear strengths. For example, T_f represents the tensile strength of carbon fibers.

It is necessary to define the material properties of each component of the composites. The 3D Hashin criteria [39], which are widely validated for various composite systems, were used to predict longitudinal tensile and compressive damage in the yarns through the user defined sub-routine UMAT. The longitudinal tensile failure coefficient F_{lt} is defined as:

$$F_{lt}^2 = \left(\frac{\sigma_{11}}{T_{y11}}\right)^2 + \frac{\sigma_{12}^2}{S_{y12}^2} + \frac{\sigma_{13}^2}{S_{y13}^2} \geq 1, (\sigma_{11} \geq 0) \quad (10)$$

The compressive tensile failure coefficient F_{lc} is:

$$F_{lc}^2 = \left(\frac{\sigma_{11}}{C_{y11}}\right)^2 + \frac{\sigma_{12}^2}{S_{y12}^2} + \frac{\sigma_{13}^2}{S_{y13}^2} \geq 1, (\sigma_{11} < 0) \quad (11)$$

The transverse and normal tensile failure coefficient is:

$$F_{2(3)t}^2 = \left(\frac{\sigma_{22} + \sigma_{33}}{T_{y22}}\right)^2 + \frac{\sigma_{23}^2 - \sigma_{33}\sigma_{22}}{S_{y23}^2} + \frac{\sigma_{12}^2}{S_{y12}^2} + \frac{\sigma_{13}^2}{S_{y13}^2} \geq 1, (\sigma_{22} + \sigma_{33} \geq 0) \quad (12)$$

$$F_{2(3)c}^2 = \frac{\sigma_{22} + \sigma_{33}}{C_{y22}} \left[\left(\frac{C_{y22}}{2S_{y23}} \right)^2 - 1 \right] + \left(\frac{\sigma_{22} + \sigma_{33}}{2S_{y23}} \right)^2 + \frac{\sigma_{23}^2 - \sigma_{33}\sigma_{22}}{S_{y23}^2} + \frac{\sigma_{12}^2}{S_{y12}^2} + \frac{\sigma_{13}^2}{S_{y13}^2} \geq 1, (\sigma_{22} + \sigma_{33} < 0)$$

The damage criterion proposed by Christensen [40] was adopted to study matrix damage, the failure coefficient F_m is:

$$F_m^2 = \left(\frac{1}{T_m} - \frac{1}{C_m} \right) (\sigma_{11} + \sigma_{22} + \sigma_{33}) + \frac{1}{T_m C_m} \left\{ \frac{1}{2} [(\sigma_{11} - \sigma_{22})^2 + (\sigma_{22} - \sigma_{33})^2 + (\sigma_{33} - \sigma_{11})^2] + 3(\sigma_{12}^2 + \sigma_{13}^2 + \sigma_{23}^2) \right\} \geq 1 \quad (13)$$

When the initial damage criteria satisfy the maximum stress criterion, the stiffness of the damaged element is reduced. The stiffness of the material is degraded as damage develops. The damage variables can be calculated as follows.

$$d_I = \min \left(1, \max \left(0, \frac{\varepsilon_I^e (\varepsilon_I - \varepsilon_I^0)}{\varepsilon_I (\varepsilon_I^e - \varepsilon_I^0)} \right) \right), I = F_{1t}, F_{1c}, F_{2t}, F_{2c}, F_{3t}, F_{3c}, F_m \quad (14)$$

where d_I , ε_I^e , ε_I , ε_I^0 are the damage variables, equivalent failure strain, equivalent strain, equivalent initial damage strain, respectively.

The stiffness reduction model developed by Murakami [41] was used to degrade the stiffness matrix of the material.

$$D = \begin{bmatrix} d_1^2 C_{11} & d_1 d_2 C_{12} & d_1 d_3 C_{13} & 0 & 0 & 0 \\ & d_2^2 C_{22} & d_2 d_3 C_{23} & 0 & 0 & 0 \\ & & d_3^2 C_{33} & 0 & 0 & 0 \\ & & & d_{12} C_{44} & 0 & 0 \\ & & & & d_{13} C_{55} & 0 \\ sym & & & & & d_{23} C_{66} \end{bmatrix} \quad (15)$$

The Duvaut-Lions regularization [42] was used to avoid convergence problems.

$$\dot{D}_i^v = \frac{1}{\delta} (D_i - D_i^v) \quad (16)$$

where δ and D_i^v are the viscosity coefficient and the normalized damage variable, respectively.

The regularized damage variable recurrence formula for the (n+1)th step is expressed as:

$$D_i^v|_{t_0+\Delta t} = \frac{\Delta t}{\delta + \Delta t} D_i|_{t_0+\Delta t} + \frac{\delta}{\delta + \Delta t} D_i|_{t_0} \quad (17)$$

The mechanical properties of the 3D braided composites are given in Table 2.

4. Theoretical model of the off-axis mechanical properties

4.1. The space angle of the yarns in the off-axis coordinate system

Due to the complex spatial distribution of the yarns, it is difficult to determine the off-axis stiffness of 3D braided composites. In the on-axis system $oxyz$, there are four spatial orientations of the yarn in the internal RVE: (γ, φ) , $(\gamma, -\varphi)$, $(-\gamma, \varphi)$, $(-\gamma, -\varphi)$ [44]. Here, φ is the angle between the yarn's projection on the $yozy$ plane and the y -axis. In order to obtain the off-axis stiffness matrix, the on-axis system is rotated along the y -axis, and the rotation angle is equal to the off-axis angle θ , as shown in Fig. 4. In this case, the problem of solving the off-axis stiffness is transformed into the positive axis stiffness in the rotated coordinate system. In the off-axis coordinate system $ox'yz'$, the spatial orientation angles of the yarns of the inner cell are (γ', φ') , $(\gamma', -\varphi')$, $(-\gamma', \varphi')$, $(-\gamma', -\varphi')$.

For a single yarn (γ, φ) , the space vector $\vec{F}$ of the yarn is $(1, \tan \gamma \cos \varphi, \tan \gamma \sin \varphi)$, the space vector of the off-axis x' -axis is $(1, \tan \theta, 0)$, and the off-axis braided angle γ' can be obtained by using the following formula

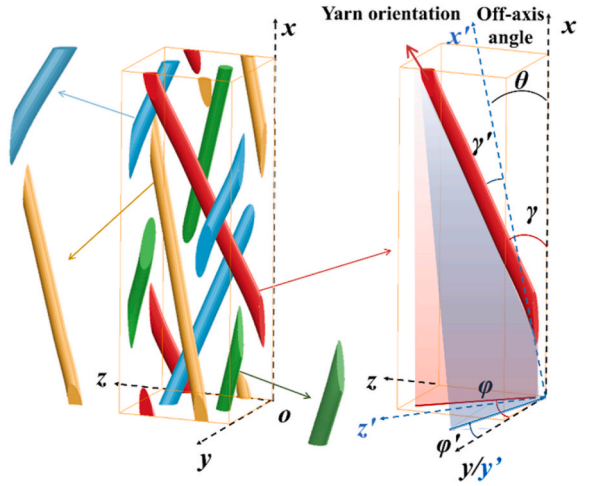


Fig. 4. The off-axis coordinate system.

under the assumption that yarns align in an idealized manner within the braided structure. This assumes that the yarns follow the geometric configuration of the braiding process without additional mechanical constraints or irregularities:

$$\gamma' = \arccos \left(\frac{\vec{F} \cdot \vec{x'}}{\|\vec{F}\|_2 \times \|\vec{x'}\|_2} \right) = \arccos \left(\frac{1 + \tan \gamma \sin \varphi \tan \theta}{\sqrt{(1 + \tan^2 \gamma)(1 + \tan^2 \theta)}} \right) \quad (18)$$

where γ' reflects the spatial orientation of the yarn in response to off-axis loading, offering a critical geometric perspective for analyzing the impact of varying off-axis angles on the braided composites' structural alignment.

The angle φ' between $\vec{T}$ (the projection vector from the yarn's space vector $\vec{F}$ to the rotated $yozy'$) and the off-axis y' space vector $(0, 1, 0)$ can be obtained by the following formula.

$$\varphi' = \text{sgn}(\varphi' - \varphi^c) \arccos \left(\frac{\vec{y'} \cdot \vec{T}}{\|\vec{y'}\|_2 \times \|\vec{T}\|_2} \right) \quad (19)$$

$$\vec{T} = \vec{F} - \vec{F} \cdot \text{norm}(\vec{x'}) \times \text{norm}(\vec{x'})$$

where sgn is the sign function, and the φ^c is the critical angle of $yozy'$ plane.

4.2. Derivation of the off-axis stiffness matrix

The stiffness matrix $[\bar{C}]$ and the flexibility matrix $[\bar{S}]$ in the global coordinate system of the RVE are given in Eq. (20) below [45–47]:

$$[\bar{C}] = \frac{1}{4} \left(\sum_{k=1}^4 [A_k][C_k][A_k]^T \right) V_y + [C]_m (1 - V_y) \quad (20)$$

$$[\bar{S}] = \frac{1}{4} \left(\sum_{k=1}^4 [B_k][S_k][B_k]^T \right) V_y + [S]_m (1 - V_y)$$

where $[C]$ and $[S]$ represent the stiffness and flexibility matrices in the

Table 2

Mechanical properties of the constituent materials of the 3D braided composites [43].

	E_{11} (GPa)	E_{22} (GPa)	ν_{12}	ν_{23}	G_{12} (GPa)	G_{23} (GPa)	T (MPa)	C (MPa)
T700-12K	230.00	18.22	0.27	0.25	36.60	7.00	4900.00	1400.00
SiC	350.00	—	0.20	—	—	—	300.00	300.00

yarn coordinate system; $[A_k]$ and $[B_k]$ are the spatial stress transformation and the strain transformation matrices, respectively.

The theoretical model proposed in this paper can be used to predict the elastic properties of 3D braided composites under off-axis loading through the mixing coefficient ρ , which improves the prediction efficiency of the model:

$$[C]_{\text{off-axis}} = \rho[\bar{C}] + (1 - \rho)[\bar{S}]$$

$$\rho = \frac{1}{1 + \tan^2 \gamma + \tan^2 \theta} \quad (21)$$

The mixing coefficient ρ is determined from the internal braid angle γ and the off-axis angle θ .

5. Results and discussion

5.1. Prediction of mechanical behavior and modulus of the 3D braided composites

Fig. 5 illustrates the stress-strain curves and strain distributions in the specimens. The on-axis (0°) specimen exhibits a linear stress-strain response, characterized by a high modulus and fiber-dominated tensile behavior. In contrast, the off-axis specimens display pronounced nonlinearity, as evidenced by continuously decreasing tangent moduli with increasing strain. Furthermore, specimens with large off-axis angles demonstrate a progressively low modulus, reflecting the influence of shear-dominated deformation mechanisms.

The gradual decrease in tensile modulus for the off-axis specimens is primarily attributed to the reorientation and deformation of mesoscale yarns. This reorientation not only alters the spatial configuration of the yarns but also shifts the load-bearing mechanism from fiber-dominated one to matrix-dominated one. As the matrix undergoes plastic deformations and interfacial sliding occurs between the yarns and the matrix, energy dissipation becomes prominent. These localized deformation mechanisms result in significant ductility, especially in the 45° specimens, despite their earlier failure compared to the 0° specimens. This behavior contrasts with the 0° specimens, where the deformation is primarily dictated by the high stiffness and load-bearing capacity of the yarns, leading to limited ductility.

Nonlinear effects are particularly evident in the 45° and 60° specimens, where a shallow slope in the stress-strain curves indicates shear-dominated deformations. The primary contribution to this nonlinearity is yarn re-orientation, which facilitates the redistribution of stress within the composites. Matrix damage further exacerbates the

degradation of load transfer efficiency between yarns, amplifying the nonlinear response.

These results highlight that under off-axis loading, the composite material sustains a higher proportion of shear forces compared to that under on-axis loading. Moreover, the combined effects of off-axis angle and braiding angle significantly influence the mechanical behavior. Changes in the spatial orientation of the yarns (γ , φ) alter the relative contributions of the yarns and the matrix in bearing the load, ultimately resulting in variation in the mechanical properties of the composite. These findings emphasize the complex interplay between yarn architecture, matrix behavior, and loading direction in determining the overall mechanical response of braided composites.

5.2. Damage evolution

Since the damage mechanisms in the 3D braided composites under off-axis loading are complex, a number of off-axis angles were considered in the simulation. Damage evolution in the on-axis tension tests, shown in Fig. 6, are discussed below. The yarns exhibit both longitudinal damage and a minor degree of transverse damage (Fig. 6a). In the case of longitudinal damage, the damage path is traverse to the entire length of the yarn, extending from the cross-sectional center to the outer edge. Additionally, damage occurs due to stress concentrations in the matrix (Fig. 6c). As stress increases, the yarn-matrix interface is fully damaged. A substantial amount of longitudinal damage and a minor amount of transverse damage occur in the transverseloading direction, with longitudinal damage representing the primary failure mode of the yarn, followed by yarn fracture (Fig. 6b and d). Ultimately, damage in the matrix intensifies, leading to fracture of the matrix (Fig. 6f). The stress-strain curve of the on-axis test exhibits a linear response with high stiffness, indicating limited contributions from matrix and transverse damages during the deformation process. This experimental observation aligns with the simulation results, where longitudinal yarn damage dominates the failure mechanism.

During the 45° off-axis testing, debonding of the interface initially occurs and the composite fails quickly. As a result of this interface failure, the stress concentration in the yarns near the contact area increases and longitudinal damage of the yarns occurs, concentrating at the intersection of the yarns, as shown in Fig. 7a. Under tension-shear coupling, the transverse damage in the yarns occurs and the level of damage is greater than that of the longitudinal damage, Fig. 7c. At the same time, the matrix damage occurs due to the presence of stress concentration in the inclined pocket of the matrix, Fig. 7e. As loading continues, transverse damage in the yarns occurs. Finally, longitudinal and transverse damage in the yarns as well as matrix damage cause complete failure of the sample. The experimental stress-strain curve for the 45° case exhibits a nonlinear behavior, consistent with the shear-dominated deformation observed in the simulations. Here, the common damage modes are interface debonding, matrix cracking, and yarn fracture.

These findings reveal that the loading direction has a significant influence on the failure mode of the yarn and the primary cause of yarn pull-out is tension-shear damage in the yarns, particularly at the intersection of the yarns.

5.3. SEM analysis of the fracture surfaces

Tensile fracture at the mesoscale and microscale of the 3D braided C/SiC composite was observed in the SEM. The fracture surface of Specimen T-0-1 exhibits a serrated appearance in Fig. 8a. Fracture of the filament occurs in a direction that is perpendicular to the direction of the filaments at the microscale, Fig. 8d. The observed damage modes were yarn fracture and filament pull-out. The matrix between the yarns has fractured, Fig. 8c. The fracture surfaces of the yarns oriented in different directions is perpendicular to the direction of those yarns, Fig. 8b. Because there is a strong bond between the yarn and the matrix, the

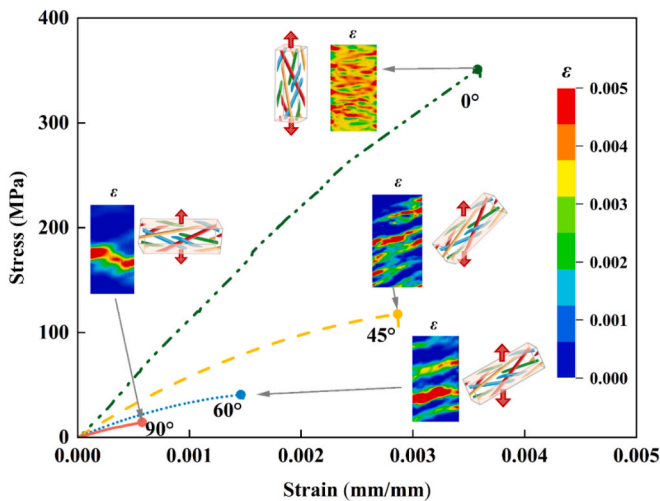


Fig. 5. Tensile stress-strain curves and strain distributions in the test specimens.

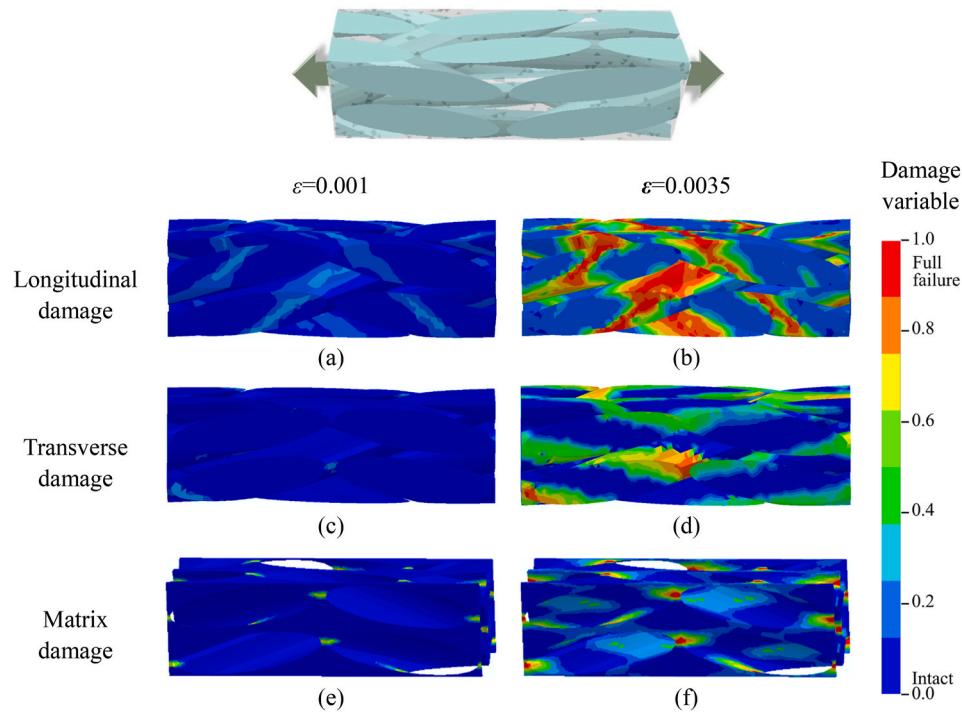


Fig. 6. Damage propagation in the braided composites subjected to the on-axis tensile loading.

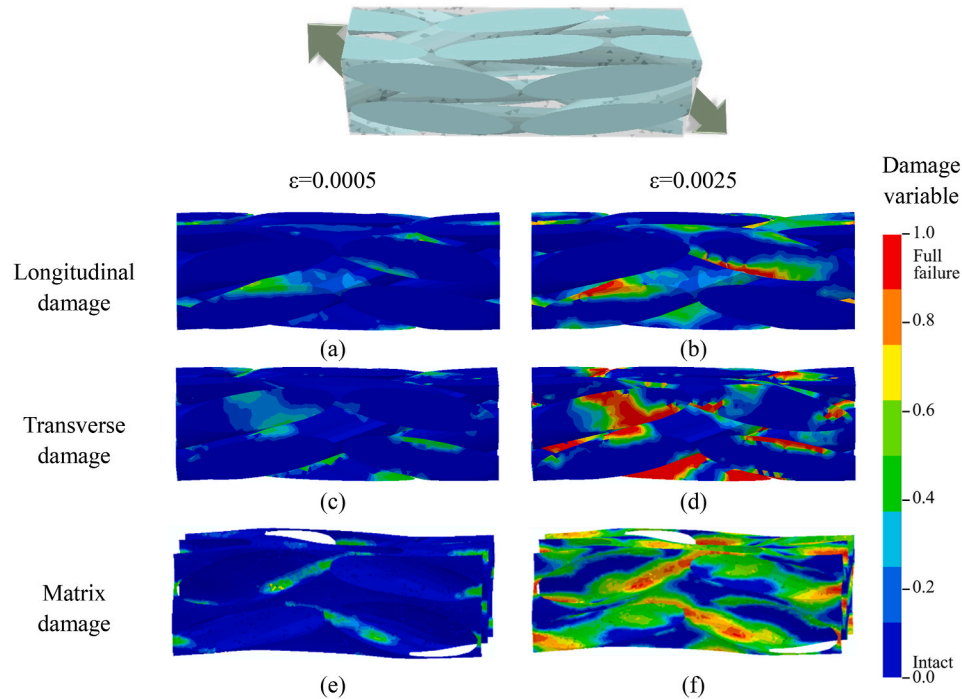


Fig. 7. Damage propagation in the braided composites subjected to the 45° off-axis tensile loading.

micro-cracks do not extend along the yarn but directly penetrates the yarn, precipitating brittle fracture of the yarn and filaments, Fig. 8e. This highlights the brittle failure mode observed in the 0° samples under tensile loading at the microscale.

The angle between the macroscopic fracture surface of Specimen T-45-1 and the tensile direction is approximately 45°, as shown in Fig. 9a. Thus, compared with Specimen T-0-1, the off-axis force applied to the yarn is increased, due to the combined effect of the off-axis angle and the braiding angle. Yarns are subjected to a tensile shear coupling force, the

fracture section of the composite is more uneven, with a number of filaments being pulled out (Fig. 9b). The filaments are damaged by the tension-shear stress field at the microscale. The angle between the fracture direction of the filaments and the tensile direction is approximately 45°, Fig. 9c. The damage morphology is shown in Fig. 9d.

The angle between the macroscopic fracture surface of Specimen T-60-1 and the tensile direction is approximately 60°, as shown in Fig. 10a. Compared with T-45-1, the angle between the off-axis angle and the spatial orientation angle of the yarn increases, the yarn cannot exert the

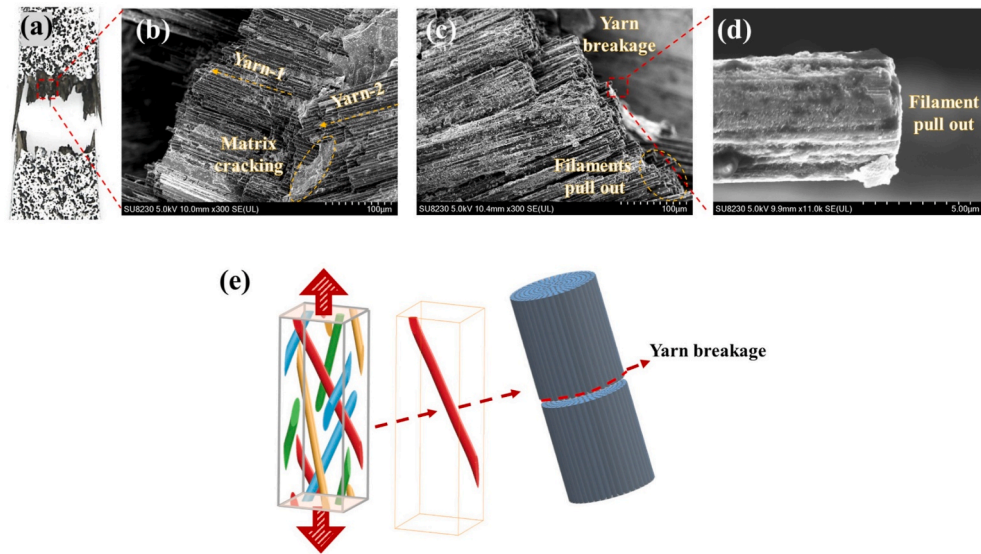


Fig. 8. Tensile fracture morphology of Specimen T-0-1. (a) Specimen fracture, (b) Matrix fracture at 100 μm scale, (c) Yarn fracture at 100 μm scale, (d) Filament fracture at a 5 μm scale, (e) Diagram of damage morphology.

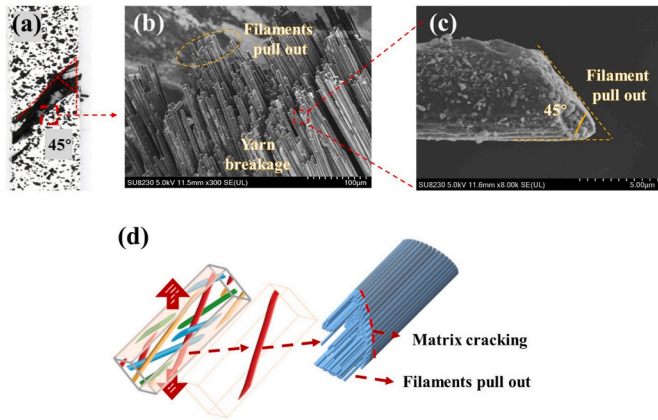


Fig. 9. Tensile fracture morphology of specimen T-45-1. (a) Specimen fracture, (b) Yarn fracture scale at 100 μm , (c) Filament fracture scale 5 μm , (d) Diagram of damage morphology.

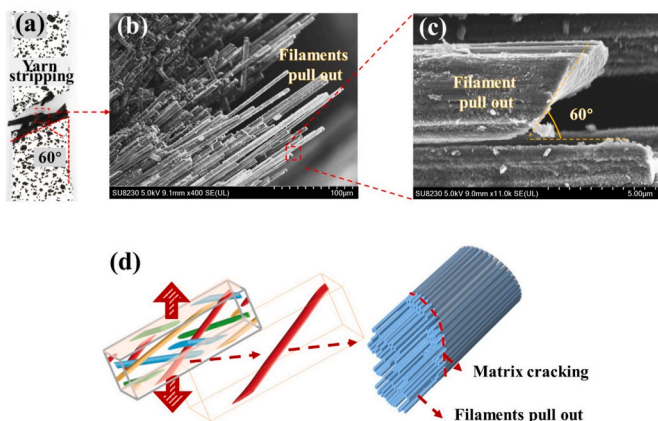


Fig. 10. Tensile fracture morphology of Specimen T-60-1. (a) Specimen fracture, (b) Yarn fracture at 100 μm scale, (c) Filament at 5 μm scale, (d) Diagram of damage morphology.

maximum bearing capacity and the matrix bears the main off-axis load. Thus, the matrix in the yarn is broken off and a small number of filaments are pulled out, Fig. 10b. The angle between the fracture of the filaments and the filaments is approximately 60° at the microscale, as shown in Fig. 10c. The damage morphology is shown in Fig. 10d.

Fig. 11a shows that the fracture surface of Specimen T-90-1 is relatively flat. The macroscopic fracture surface is perpendicular to the tensile direction. Fracture of the filaments is perpendicular to their direction at the microscale, Fig. 11d. Due to the transverse tension of the composite material under 90° loading, when the matrix is subjected to a tension load, the weak interface between yarns causes yarn stripping, with filaments being pulled out, and the matrix breaking perpendicular to the direction of the tension load, as shown in Fig. 11c and b. The damage morphology is shown in Fig. 11d.

5.4. Model prediction and design guidance

Regarding the 3D braided composites with a braiding angle of 20° , the values of γ' and ϕ' decrease continuously with increasing off-axis angle during Stage I, as shown in Fig. 12a. The green dots in Fig. 12b represent the results predicted by the off-axis stiffness model in Section 4.1. As the off-axis angle gradually approaches the braiding angle of the internal yarns, the shear force on the composite decreases, and the excellent axial bearing capacity of the yarns becomes apparent, resulting in a gradual increase in the modulus of the composites. In Stage II, with increasing off-axis angle, the values of γ' and ϕ' also increase. As the angle between the applied tension and the yarns increases, there is an increase in the shear force applied to the composite. This evidence suggests that the simulations capture the experimental results accurately. Under combined tension-shear off-axis loading, the composites fail prematurely, due to their low interfacial strength. To a certain extent, increasing the off-axis angle will improve the modulus of the braided composite, and reasonable design the off-axis angle of the 3D braided composite structure can maximize the use of the excellent characteristics of the composite. This finding provides a reference for the design of braided structures.

While the combined experimental, theoretical, and multiscale modeling approach provides valuable insights into the behavior of 3D braided C/SiC composites, it has certain limitations. The theoretical model assumes idealized material properties and perfect yarn-matrix interfaces, which may not fully capture the interfacial debonding. Additionally, the finite element model involves high computational

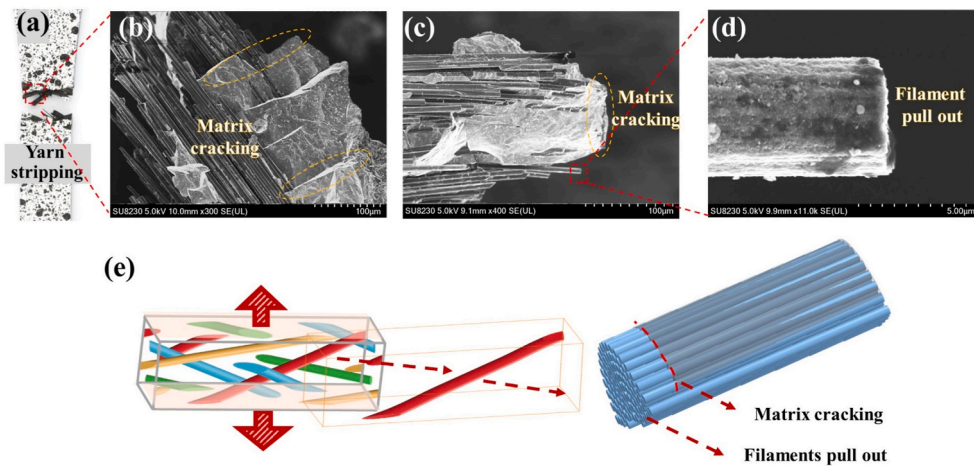


Fig. 11. Tensile fracture morphology of Specimen T-90-1. (a) Specimen fracture, (b) Matrix fracture at 100 μm scale, (c) Yarn fracture at 100 μm scale, (d) Filament fracture at a 5 μm scale, (e) Diagram of damage morphology.

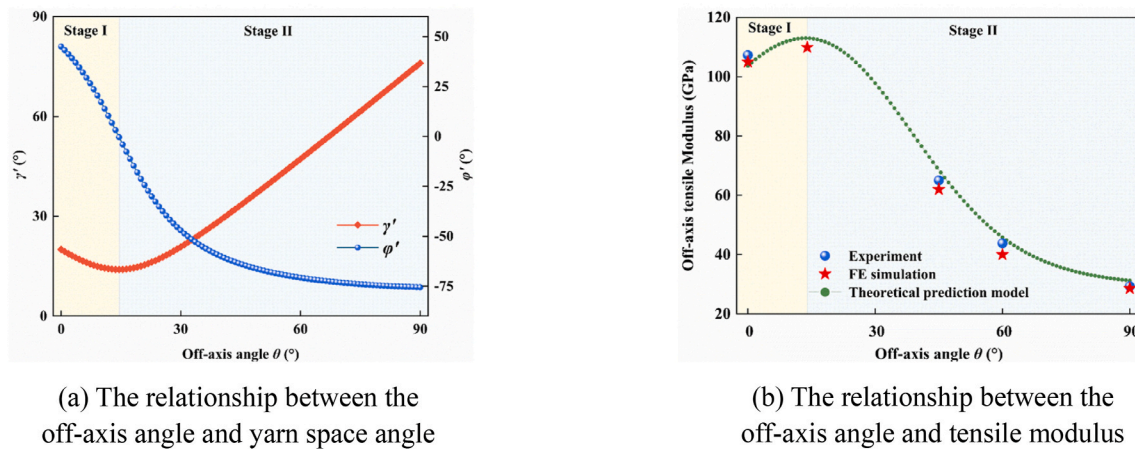


Fig. 12. Influence of off-axis angle on the space angle of the yarn and the off-axis tensile modulus.

costs due to the intricate details of the braided structure. Future research may need to focus on developing more efficient algorithms to reduce computational cost.

6. Conclusions

A combined experimental, theoretical, and multiscale modeling approach has been conducted to investigate the strength and failure behavior of 3D braided C/SiC composites under off-axis tensile loading. A theoretical model has been developed to predict the off-axis modulus of the composites. The influence of varying the off-axis angle on the spatial orientation angle of the braided yarns and the off-axis tensile modulus has been investigated. An FE model has been used to simulate the progressive failure process in the braided composites under the off-axis loading. The FE model has revealed the detailed damage evolution processes and the failure mechanisms in both the matrix and the yarns. The numerically-predicted stress-strain responses and failure mechanisms have been shown to be in good agreement with the corresponding experimental results. The SEM technique has been used to investigate the fracture surfaces of the test specimens and the identify the failure mechanisms under various off-axis loading conditions. The findings indicate that as the off-axis loading angle increases, the spatial orientation angle of the composite yarn changes, and the off-axis stiffness of the braided composite initially increases and then decreases. Moreover, significant localized ductility is observed under off-axis tensile loading.

This ductility is primarily attributed to the interfacial sliding between yarns and the matrix, as well as yarn reorientation, which accommodate local strains and delay complete failure in certain regions. Under off-axis loading, the shear-dominated deformation mechanism accelerates damage initiation and propagation, resulting in earlier failure compared to the 0° specimens. Despite this, the shear mechanism promotes interfacial sliding and stress redistribution, enhancing the localized strain capacity of the material, particularly in the 45° specimens. These findings emphasize the complex interplay of yarn structure, matrix brittleness, and interfacial behavior under various loading directions, providing critical insights for optimizing the mechanical performance of ceramic matrix composites in practical applications.

CRediT authorship contribution statement

Xinyi Song: Writing – original draft, Software, Methodology, Data curation, Conceptualization. **Jin Zhou:** Writing – review & editing, Supervision, Project administration, Conceptualization. **Jun Wang:** Resources, Project administration. **Longteng Bai:** Resources. **Xiaohui Yang:** Investigation. **Jie Xue:** Resources, Investigation. **Di Zhang:** Resources, Methodology, Formal analysis. **Shenghao Zhang:** Methodology, Investigation. **Xuefeng Chen:** Writing – review & editing, Methodology. **Zhongwei Guan:** Writing – review & editing, Methodology. **Wesley J. Cantwell:** Writing – review & editing, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This research was funded by Natural Science Basic Research Program of Shaanxi [grant number 2023-JC-QN-0042] and the National Key R&D Program of China [grant number 2021YFF0500100].

Data availability

Data will be made available on request.

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